

Dual-Memory Dissociation Under Partial Observability: An Identifiability-Governed Phase Transition

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Abstract

We study whether two latent memory channels — an E-channel driven by fast, high-contrast binary growth-rate switches (g_{latent}) and a B-channel driven by slow capacity drift (C_{eff}) — produce a statistically detectable double dissociation in the lifecycle model output, and how this detection degrades as oracle feature quality is reduced. Using a controlled simulation framework with three observation worlds (World C: informative sensor; World Hybrid: L/C sensor with empirical E-channel blindness under tested decoders; World A2: change-labeled null control), we demonstrate four findings. **Phase 1** (oracle feature degradation, $N = 50$ seeds): (1) *asymmetric channel damage* — the E-channel metric D_{rate} maintains 100% pass rate down to oracle $R_g^2 = 0.028$ (masking 20% of anchors) while the B-channel metric D_{MB} collapses at the same level, because the two channels require different information structures (binary contrast vs. continuous trend tracking); (2) per-channel recoverability (R_g^2 and R_C^2 , each an operational recoverability proxy on held-out data) is necessary to diagnose this asymmetry, which is invisible in the scalar average R^2 . **Phase 2** (true partial observability: sensor degradation + Ridge decoder, $N = 50$ seeds): (3) E-channel *recoverability* governs a transition zone at decoder $R_g^2 \approx 0.06\text{--}0.15$ — above this band D_{rate} reliably exceeds chance; below it, D_{rate} becomes indistinguishable from chance across tested decoders; this transition is *decoder-robust* across five Ridge regularization strengths and an MLP. Note: the Phase 1 floor ($R_g^2 = 0.028$) is not the Phase 2 operational limit; the two phases differ in observation operator, decoder, and degradation mechanism. (4) World Hybrid establishes an empirical non-recoverability baseline: when $X = f(L/C)$, g_{latent} is unrecoverable from X across tested decoders (Ridge and MLP), so D_{rate} is at chance across all sensor levels; D_{MB} remains above chance because C_{eff} leaks through the L/C denominator by design. All results are validated by a comprehensive skeptic suite (label shuffle, permuted retrain, feature permutation, column swap, exact recompute). This paper is an existence proof within defined synthetic

generative worlds: the reported patterns constitute a recoverability-governed transition under partial observability, scoped to the defined generator and observation families.

Keywords: Simulation modelling; Partial observability; Identifiability threshold; Latent state inference; Regime switching models; Inverse problems; Decoder robustness; Computational experiments

Introduction

Economic lifecycle models routinely postulate that multiple distinct latent memory channels govern output dynamics — for example, an entrepreneur’s current productivity level (g -channel) versus the accumulated capacity stock (C -channel). A key empirical question is whether these channels are *non-substitutable*: does removing knowledge of one channel specifically impair prediction of that channel’s associated outcomes while leaving the other channel’s predictions intact? This double dissociation structure is well-established in cognitive neuroscience as a criterion for functional independence (Teuber 1955; Shallice 1988), but has rarely been applied formally to economic latent-variable models. Motivated by the TFP framework’s hypothesis that lifecycle systems require two distinct memory channels, we construct controlled generative worlds to test whether causal attribution exhibits a double dissociation under partial observability.

We address two intertwined questions:

1. *Does a double dissociation exist at full observability (oracle)?* That is, when we have perfect knowledge of both latent channels, can we show that each channel selectively contributes to predicting its associated regime?
2. *How robust is this dissociation to degraded observability?* As sensor quality declines and we must use learned decoders to reconstruct latent signals, at what recoverability threshold does the dissociation collapse?

Structural identifiability vs. operational recoverability.

We use two distinct concepts throughout this paper and apply them to different phases. *Structural identifiability* (Bellman and Åström 1970; Rothenberg 1971; Kalman 1960) asks whether the mapping from parameters to observables is injective in principle, assuming perfect data and unlimited sample size. *Practical identifiability* (Raue et al. 2009) asks whether finite, noisy data permit estimation of structurally identifiable parameters with acceptable precision. Both classical concepts are grounded in the tradition of linear systems theory and observability (Kalman 1960) and in the theory of ill-posed inverse problems (Tikhonov and Arsenin 1977; Engl et al. 1996), which formalizes when latent states cannot be recovered from observations regardless of estimation strategy.

Our Phase 1 and Phase 2 analyses use a third, operational concept: *recoverability*. A latent channel is *recoverable* if a decoder trained on the observed sensor X_t achieves $R^2 > 0$ on held-out test data. This is a strictly empirical criterion — a property of the combination of observation operator, decoder family, and data regime — not a claim about the underlying

parameter space. We quantify E-channel recoverability as R_g^2 (decoder R^2 for g_{latent}) and B-channel recoverability as R_C^2 (decoder R^2 for C_{eff}), both on held-out test series. In Phase 2, the term *identifiability* is reserved for the formal structural sense; all operational measurements are reported as *recoverability*.

Our setting also admits an empirical non-recoverability finding: when $X_t = f(L_t/C_{\text{nominal}})$, the E-channel is unrecoverable from X under tested level-based decoders, consistent with a dilution argument (Section 3.3.2). This three-way distinction organizes our experimental design: Phase 1 explores recoverability degradation under oracle-feature perturbations; Phase 2 explores sensor-imposed recoverability limits under noise; World Hybrid instantiates the empirical non-recoverability baseline (under tested level-based decoders).

Our main contribution is a *phase transition characterization* of the dissociation’s recoverability: we show that the two channels have asymmetric recoverability thresholds, mechanistically explained by the fast/high-contrast vs. slow/low-contrast nature of their latent signals. We further demonstrate that this transition zone is *decoder-robust* — stable across regularization strengths and decoder types within the tested configurations — and provide an empirical non-recoverability finding for observation families that structurally dilute E-channel information (under tested level-based decoders).

Companion empirical manuscript.

While this manuscript is strictly synthetic, the dissociation test and lightweight memory-proxy framing have also been instantiated in an empirical degradation setting (lithium-ion batteries) in a companion manuscript (Ollar 2026). That empirical study is separate in scope and does not validate the specific generator families analyzed here; it is cited only to demonstrate that the dissociation test can be posed and evaluated in real, partially observed systems under genuine sensor constraints.

Methods

Generator Worlds

We work with three simulation worlds, all built on a common lifecycle model with a grow-or-maintain production function. Let L_t be the load at time t , g_t the latent growth rate drawn from $\{g_{\text{lo}}, g_{\text{hi}}\}$ via a Markov chain with switch probability $p_g = 0.05$, and $C_{\text{eff},t}$ the effective capacity subject to exogenous fatigue shocks. Regime labels are assigned per cycle based on the prevailing latent states.

World C. Sensor observation $X_t = \tanh(W \cdot [L_t, C_t, g_t, k_t]^T + \varepsilon_t)$ where W is a fixed random mixing matrix. This sensor directly encodes g_{latent} and C_{eff} in the mixing, making both channels decodable via a sufficiently powerful regressor. Train/validation/test split: 72/24/24 series out of 120 total; decoder fitted on TRAIN only; R^2 evaluated on TEST.

World Hybrid. Sensor $X_t = f(\varphi_t)$ where $\varphi_t = L_t/C_{\text{nominal}}$ (load-to-capacity ratio). The E-channel information is *empirically unrecoverable* from X_t under tested level-based decoders (dilution argument, Section 3.3.2).

World A2 (null control). Regime labels are assigned based on *change events* (S_E switched this cycle) rather than latent state levels. Oracle features encode level-based signals, which are orthogonal to change-event labels by construction. World A2 provides the correct null: if the dissociation test passes for A2, it would indicate a pipeline artifact, not a structural finding.

Dissociation Metrics

Let $\text{AUROC}(y, \hat{p}; \text{cls})$ denote the one-vs-rest AUROC for class cls. Define:

$$\begin{aligned} D_{\text{rate}} &= [\text{AUROC}(\text{full}, E\text{-only}) - \text{AUROC}(M_C\text{-only}, E\text{-only})] \\ &\quad - [\text{AUROC}(\text{full}, B\text{-only}) - \text{AUROC}(M_C\text{-only}, B\text{-only})] \\ D_{\text{MB}} &= [\text{AUROC}(\text{full}, B\text{-only}) - \text{AUROC}(M_g\text{-only}, B\text{-only})] \\ &\quad - [\text{AUROC}(\text{full}, E\text{-only}) - \text{AUROC}(M_g\text{-only}, E\text{-only})] \end{aligned}$$

where M_C -only ablates the g -channel feature and M_g -only ablates the C -channel feature. $D_{\text{rate}} > 0$ indicates the g -channel selectively contributes to E-only detection; $D_{\text{MB}} > 0$ indicates the C -channel selectively contributes to B-only detection. Pass rate is the fraction of seeds with $D > 0$.

Phase 1: Oracle Feature Degradation Ladder

Oracle features are exponential moving averages (EMA, $\alpha = 0.02$) of the true latent signals: $[\text{EMA}(g_t), \text{EMA}(C_{\text{eff},t})]$. We apply $K = 7$ degradation levels ($L0$: oracle; $L1$: noise $\sigma = 0.1 \cdot \text{sd}$; $L2$: 16-bin quantize; $L3$: noise $\sigma = 0.5 \cdot \text{sd}$; $L4$: mask 20%; $L5$: noise $\sigma = 2 \cdot \text{sd}$; $L6$: blind) and measure per-seed D_{rate} and D_{MB} across $N = 50$ seeds.

Per-channel recoverability is measured separately: $R_g^2 = R^2(\text{degraded EMA}(g), \text{oracle EMA}(g))$ and analogously R_C^2 . The 2D logistic classifier uses $[R_g^2, R_C^2]$ jointly to predict pass/fail.

Phase 2: Sensor Degradation, Decoder, and E-Channel Recoverability

We add Gaussian noise to the observed sensor X at five levels ($\sigma \in \{0, 0.5, 1.0, 2.0, 5.0\} \cdot \text{sd}_X$, where sd_X is the training-set standard deviation). A Ridge decoder ($\alpha = 1.0$) is fitted separately on TRAIN-degraded data to predict g_{latent} and $C_{\text{eff}, \text{latent}}$ from X_{deg} . The decoded features are used as oracle-feature substitutes at anchor points. $N = 50$ seeds; all normalization uses train-set statistics only. We operationalize *recoverability* as latent decodability under the observation operator and decoder family: a channel is recoverable if a decoder trained on X_t achieves $R_g^2 > 0$ on held-out test series. R_g^2 is thus an operational recoverability proxy, not a claim of strict parameter identifiability in the classical structural sense (Bellman and Åström 1970; Rothenberg 1971).

Decoder Robustness Checks

To confirm that Phase 2 results are not artifacts of the Ridge configuration, we run:

- **Ridge alpha sweep:** $\alpha \in \{0.01, 0.1, 1.0, 10, 100\}$, $N = 10$ seeds, same protocol as Phase 2.

- **MLP comparison:** MLPRegressor(32, 16), ReLU, Adam, early stopping on held-out 10% of training set, subsampled to 20K training points, $N = 3$ seeds (small confirmatory sample), $L0$ – $L4$ levels.

Skeptic Validation Suite

All oracle results are validated by:

Tier A

Label shuffle null: 2000 permutations of saved predictions. AUROC should approach 0.50.

Tier B

Permuted retrain: 125 full retrains with permuted labels. Perm AUROC should be ≈ 0.50 .

Test B

Feature permutation: within-seed shuffling of feature columns. AUROC should collapse to 0.50.

Test C

Column swap + sign flip: swap M_E and M_B features; D_{rate} and D_{MB} should flip sign.

Test D

Train-test gap: $|\text{AUROC}_{\text{train}} - \text{AUROC}_{\text{test}}| < 0.002$ (no leakage).

Test E

Independent recompute: re-compute D from saved predictions without retraining. Max $|\Delta| = 0$.

Results

Oracle Baseline (Figure 1)

At full observability ($N = 50$ seeds each), state-labeled worlds (C, Hybrid) achieve strong double dissociation; World A2 (null control) is near-chance throughout (Table 1).

Oracle baseline results ($N = 50$ seeds). CI = 95% bootstrap confidence interval.

World	D_{rate} [95% CI]	Pass	D_{MB} [95% CI]	Pass	E-AUROC
A2 (null)	-0.0084 [$-0.023, +0.006$]	48%	$+0.0063$ [$-0.000, +0.013$]	62%	0.515

World	D_{rate} [95% CI]	Pass	D_{MB} [95% CI]	Pass	E-AUROC
C	+0.1628 [+0.153, +0.173]	100%	+0.0239 [+0.021, +0.027]	98%	0.859
Hybrid	+0.1716 [+0.160, +0.183]	100%	+0.0202 [+0.017, +0.024]	96%	0.860

All skeptic checks pass: Tier A null AUROC $\approx$ 0.500, Tier B perm AUROC 0.489–0.509, Test C sign flip 96–100% of seeds, Test E max $|\Delta| = 0$.

Phase 1: Asymmetric Channel Damage (Figure 2)

Table 2 shows the Phase 1 ladder results for World C. The key finding is mechanistically asymmetric channel damage at $L4$ (20% anchor masking):

Phase 1 observability ladder, World C ($N = 50$ seeds, $K = 7$ levels).

Level	Degradation	R_g^2	R_C^2	D_{rate} Pass	D_{MB} Pass
L0	Oracle	1.000	1.000	100%	98%
L1	Noise 0.1σ	0.990	0.990	100%	98%
L2	Quantize 16-bin	0.988	0.988	100%	98%
L3	Noise 0.5σ	0.802	0.792	100%	94%
L4	Mask 20%	0.028	0.288	100%	48%
L5	Noise 2.0σ	0.204	0.189	98%	78%
L6	Blind	0.001	0.001	48%	42%

Key finding: Asymmetric channel damage at $L4$. Masking 20% of anchors to zero collapses R_g^2 to 0.028 (catastrophic, due to large mean-shift bias in the high-mean, low-variance EMA of g_{latent}) while R_C^2 remains 0.288. Despite this, D_{rate} holds at 100% pass — the E-channel dissociation test survives because the 80% of intact anchors provide a high-contrast binary signal. D_{MB} collapses to 48% because the B-channel trend-tracking task requires *continuous* anchor coverage; 20% punctuated zeroing disrupts the trajectory estimate more severely than equivalent additive noise.

This asymmetry is not visible in scalar $R_{\text{avg}}^2 = 0.158$ at $L4$. The per-channel view (R_g^2, R_C^2) is essential for diagnosing it.

Phase 2: Recoverability and Empirical Non-Recoverability (Figure 3)

World C: E-Channel Recoverability Governs the Transition Zone

Table 3 shows the Phase 2 results for World C ($N = 50$ seeds).

Phase 2 sensor degradation ladder, World C ($N = 50$ seeds). Ridge decoder, train-only fit.

Level	σ_{noise}	R_g^2	R_C^2	D_{rate}	Pass	D_{MB} Pass
L0	0 (clean)	0.398	0.708	+0.037	86%	82%
L1	$0.5 \cdot \text{sd}_X$	0.263	0.566	+0.022	76%	90%
L2	$1.0 \cdot \text{sd}_X$	0.149	0.371	+0.009	64%	80%
L3	$2.0 \cdot \text{sd}_X$	0.061	0.153	+0.005	60%	66%
L4	$5.0 \cdot \text{sd}_X$	0.012	0.009	+0.006	60% [†]	52% [†]

[†]Not statistically distinct from chance ($p > 0.15$, binomial).

At clean sensor (L0), D_{rate} passes in 86% of seeds (43/50; 95% CI [73%, 94%], Clopper-Pearson; vs. 100% oracle); the gap reflects the Ridge decoder introducing $\sim 4 \times$ more noise than the oracle EMA, with per-seed D_{rate} values ranging $[-0.029, +0.171]$ (mean +0.037 vs. oracle mean +0.163). The transition zone in E-channel *recoverability* sits at $R_g^2 \approx 0.06$ – 0.15 (L2–L3): above this band D_{rate} reliably exceeds chance; below it, E-channel information is insufficient across tested decoders (Ridge and MLP).

World Hybrid: E-Channel Blindness Under Tested Decoders

Phase 2 sensor degradation ladder, World Hybrid ($N = 50$ seeds). All D_{rate} values at chance.

Level	σ_{noise}	R_g^2	R_C^2	D_{rate} Pass	D_{MB} Pass
L0	0 (clean)	−0.000	0.342	56% [†]	66%
L1	$0.5 \cdot \text{sd}_X$	−0.000	0.307	52% [†]	52% [†]
L2	$1.0 \cdot \text{sd}_X$	−0.000	0.244	30% [†]	82%
L3	$2.0 \cdot \text{sd}_X$	−0.000	0.132	40% [†]	70%
L4	$5.0 \cdot \text{sd}_X$	−0.000	0.011	48% [†]	64% [†]

Empirical finding (tested decoders). Under the L/C observation operator, both Ridge ($\alpha = 1.0$, $N = 50$ seeds) and MLP (32+16 hidden, $N = 3$ seeds) decoders achieve $R_g^2 \approx -0.000$ at all five noise levels (Table 4). This is consistent with a *dilution argument* (Tikhonov and Arsenin 1977): $X_t = f(L_t/C_{\text{nominal}})$ where $L_t = \sum_{s=1}^t [g_s \cdot g_{\text{base}} + \varepsilon_s]$ accumulates all past growth rates. Since $\text{Corr}(g_t, L_t) \rightarrow 0$ as $t \rightarrow \infty$ (one step’s contribution diluted by t steps of history), a level-based decoder on X_t cannot isolate g_t .

Scope caveat. We do not claim this holds for all conceivable decoder architectures. Increment-based or time-difference decoders applied to ΔX_t are untested and may recover partial g_t signal. The claim is strictly: under the tested Ridge and MLP decoders operating on X_t levels, E-channel recoverability is empirically zero. This is sufficient to establish World Hybrid as a diagnostic baseline for the tested pipeline.

Consequence. Because $R_g^2 \approx 0$ across all tested decoders and all degradation levels, the E-channel features passed to the LR classifier are uninformative, making D_{rate} empirically bounded at chance. This is not a general claim about decoder-class impossibility — it is an empirical non-recoverability result under the tested level-based decoding approach, consistent with the dilution argument above.

Hybrid D_{MB} : designed selectivity, not a failure. World Hybrid is an empirical non-recoverability baseline *only for E-channel recoverability* under the L/C observation operator (tested level-based decoders). It is *not* a non-recoverability baseline for the B-channel. C_{eff} (the B-channel) directly influences the L/C ratio through the denominator: $\varphi_t = L_t/C_{\text{nominal}}$. As a result, B-channel information *leaks* through the observation, making C_{eff} partially recoverable ($R_C^2 = 0.342$ at clean sensor). Accordingly, D_{MB} remains above chance in Phase 2 (Table 4: 52–82% pass across noise levels; $R_C^2 = 0.342$ at $L0$ declining to 0.011 at $L4$). This is by design: the world was constructed to isolate E-channel non-recoverability (under tested level-based decoders) while leaving the B-channel observable. That D_{MB} is above chance in Hybrid is a *confirmation*, not a contradiction, of the structural logic.

Decoder Robustness (Core Result)

Hyperparameter invariance. Table 5 shows D_{rate} pass rates across five Ridge regularization strengths ($N = 10$ seeds each). Pass rates are *identical* for $\alpha \in \{0.01, 0.1, 1.0, 10\}$ at every degradation level; only $\alpha = 100$ diverges by 10% at $L3$.

Ridge alpha sweep: D_{rate} pass rate by level and α ($N = 10$ seeds, World C).

Level	R_g^2 ($\alpha=1$)	$\alpha=0.01$	$\alpha=0.1$	$\alpha=1.0$	$\alpha=10$	$\alpha=100$
L0	0.458	100%	100%	100%	100%	100%
L1	0.318	100%	100%	100%	100%	100%
L2	0.196	70%	70%	70%	70%	70%
L3	0.087	60%	60%	60%	60%	50%
L4	0.018	50%	50%	50%	50%	50%

Decoder robustness (Ridge and MLP). The MLP decoder (32+16 hidden, early stopping, $N = 3$ seeds, 20K subsampled training points) achieves $R_g^2 = 0.638$ at clean sensor vs. Ridge $R_g^2 = 0.458$ ($N = 10$ seeds), confirming exploitable nonlinear structure in the World C sensor. MLP pass rates: 100% at $L0$ – $L1$, dropping to 67% (2/3 seeds) at $L2$ – $L4$; Ridge ($N=10$) rates: 100%, 70%, 60%, 50% at $L1$ – $L4$ respectively. Both decoders show transition onset at $L2$ (around $R_g^2 \approx 0.10$ – 0.20), consistent with the transition zone identified in the main $N = 50$ Ridge run. Note: with only $N = 3$ MLP seeds, pass rates at $L2$, $L3$, and $L4$ are indistinguishable (2/3 each); the MLP run is therefore treated as qualitative confirmatory evidence, not a precise characterization of the transition boundary.

Synthesis. The transition zone at $R_g^2 \approx 0.06\text{--}0.15$ is not a Ridge artifact. It reflects the minimum decodable E-channel information required for the dissociation classifier to distinguish E-only regimes — a structural property of the latent signal, stable across tested decoders (Ridge and MLP).

Discussion

Why the Two Channels Have Asymmetric Thresholds

D_{rate} (E-channel) uses g_{latent} , which switches between two discrete states with a high-contrast ratio ($g_{\text{hi}}/g_{\text{lo}}$ large) and a fast switch rate ($p_g = 0.05$). Any small amount of decodable information suffices: even at $R_g^2 = 0.028$ (Phase 1 $L4$, masking), 80% of intact anchors provide a strong enough binary contrast for the LR classifier. D_{MB} (B-channel) uses C_{eff} , a slow, low-contrast continuous drift. This task requires *tracking a trend* across consecutive anchor points; structured information loss (masking) that removes 20% of anchors disrupts the trend more severely than adding noise to all anchors.

The Phase 1 result that $L4$ (masked, $R_g^2 = 0.028$, $R_C^2 = 0.288$) collapses D_{MB} to 48% while $L5$ (noisy, $R_g^2 = 0.204$, $R_C^2 = 0.189$) achieves $D_{\text{MB}} = 78\%$ — despite $L4$ having *higher* R_C^2 — directly illustrates this mechanism. The two channels require different information structures, not just different information quantities.

This asymmetric per-channel recoverability is the mechanistic key: at $L4$, $R_g^2 = 0.028$ while $R_C^2 = 0.288$ — a factor-of-ten gap. E-channel detection (D_{rate}) survives because high-contrast binary signals are robust to partial loss; B-channel tracking (D_{MB}) collapses because trend reconstruction requires structurally intact consecutive anchors. Per-channel recoverability (R_g^2 , R_C^2) is therefore necessary to diagnose *why* one channel collapses before the other — scalar R_{avg}^2 obscures this asymmetry entirely.

Scope and Limits

This paper claims:

1. Within the defined generator/observation families, the dissociation test (D_{rate} , D_{MB}) detects dual-memory structure at oracle recoverability ($R^2 = 1$) and degrades through a recoverability-governed phase transition.
2. The phase transition persists under true partial observability (sensor + Ridge decoder) for World C, and is stable across tested decoders (Ridge, MLP) and regularization strengths.
3. The per-channel view (R_g^2 , R_C^2) is necessary to diagnose asymmetric channel damage that scalar R_{avg}^2 obscures.

4. World Hybrid provides an empirical non-recoverability baseline: under tested Ridge and MLP decoders on X_t levels, E-channel $R_g^2 \approx 0$ at all noise levels, consistent with a dilution argument. D_{rate} remains at chance throughout.

This paper does *not* claim:

- **Universality.** Recoverability thresholds ($R_g^2 \approx 0.06\text{--}0.15$ for the D_{rate} transition zone; $R_{\text{avg}}^2 \approx 0.15$ for D_{MB}) are world-specific and depend on generative parameters (p_g, p_k , signal contrast, cycle length). They should not be treated as universal cutoffs applicable beyond this generator family.
- **Causality.** The dissociation test is a statistical fingerprint of the generative process; it does not identify individual causal pathways.
- **Sensor-regime generality.** Phase 2 results are for World C’s default sensor. Different sensor designs may make channels structurally unobservable independently of degradation level.
- **Applicability beyond simulation.** All results are from known-ground-truth simulation. Application to real TFP data requires validation of the world model, confirmation that real sensors provide decodable latent signals, and assessment of appropriate regime labeling.

Interpretation within the TFP Framework (Q4A/Q4B Mapping)

Within the TFP framework, we interpret the rate-memory channel (g_{latent} , fast binary switching) and the capacity-memory channel (C_{eff} , slow drift) as operational analogs of Q4A and Q4B, respectively. Paper A does not claim universality of this mapping, but provides a rigorous existence proof, in a defined model class, that two non-substitutable lifecycle memories can be structurally necessary for attribution, and that recoverability undergoes an identifiability-governed transition under partial observability.

What is validated. The necessity of two non-substitutable lifecycle memory channels within this model class, and the existence of a sensor-quality transition zone below which the dissociation test cannot reliably detect them.

What is not validated. Universality of these thresholds across domains, across real sensors, across biological systems, or across TFP implementations with different generative mechanisms. Empirical application to real TFP data requires validation of the world model and confirmation that real sensors provide decodable latent signals. Those extensions are deferred to future work.

Conclusion

We have demonstrated that dual-memory dissociation in lifecycle models follows a recoverability-governed phase transition with mechanistically asymmetric thresholds for

the two channels. The dissociation is stable across tested decoders (Ridge, MLP) and regularization strengths, providing evidence within this model class that the transition reflects the latent signal structure rather than modeling artifacts. World Hybrid’s empirical non-recoverability result — confirmed across tested Ridge and MLP decoders — provides a clean diagnostic baseline: the same pipeline applied to a sensor that structurally dilutes E-channel information produces D_{rate} at chance across all noise levels and all tested decoders, consistent with the dilution argument.

These results lay the groundwork for applying the dual-memory dissociation test to real lifecycle data. They demonstrate that detection performance degrades through a transition zone once decoder R_g^2 enters approximately 0.06–0.15, and becomes indistinguishable from chance under extreme degradation ($R_g^2 < 0.07$) — a property of this generator and sensor family that is stable across tested decoder types and regularization settings. This transition zone, not any single threshold, is the operative limit on reliable dissociation inference within this simulation framework.

Scope. All reported results are existence proofs within defined synthetic generative worlds and observation families. The recoverability thresholds, transition zone boundaries, and channel asymmetries reported here are properties of this specific generator; they do not constitute universality claims and should not be applied to other sensor designs, generator families, or real-world data without independent validation. Any real-world claims require independent validation; see the companion battery manuscript (Ollar 2026) for an example of an empirical application of the dissociation test under real observability constraints.

CRedit Author Statement

Austin Ollar: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Project administration.

Competing Interests

The author declares no competing interests.

Data and Code Availability

All simulation code, raw results, and figure-generation scripts are included in the supplementary reproduction package (see REPRO.md). No external datasets were used in this manuscript.

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The broader TFP lifecycle framework was originally motivated by observations from the Objective Personality System developed by Dave Powers and Shan Renee. Their work is

cited here solely as attribution of inspiration; all empirical claims, test designs, and conclusions rest exclusively on the synthetic analyses reported in this manuscript.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work the author used Claude (Anthropic) to assist with code implementation, editing clarity, and consistency checking. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Skeptic Check Results

Full skeptic check results. All checks pass for state-labeled worlds (C, Hybrid).

Check	Result
Tier A null (label shuffle)	AUROC $\approx$ 0.500, $D_{\text{rate}} \approx 0$ for all worlds
Tier B permuted retrain (125 fits)	Perm AUROC = 0.489–0.509
Test B feature permutation	AUROC collapses to 0.500
Test C column swap + sign flip	96–100% of seeds for C and Hybrid
Test D train-test gap	< 0.002 (no leakage)
Test E independent recompute	Max $ \Delta = 0.00 \times 10^0$ (exact match)
Ladder L6 null (blind)	$D_{\text{rate}} / D_{\text{MB}}$ at chance (48–54% pass)
Phase 2 L4 null ($R^2 \approx 0.01$)	$D_{\text{rate}} / D_{\text{MB}}$ at chance ($p > 0.15$)

Figures

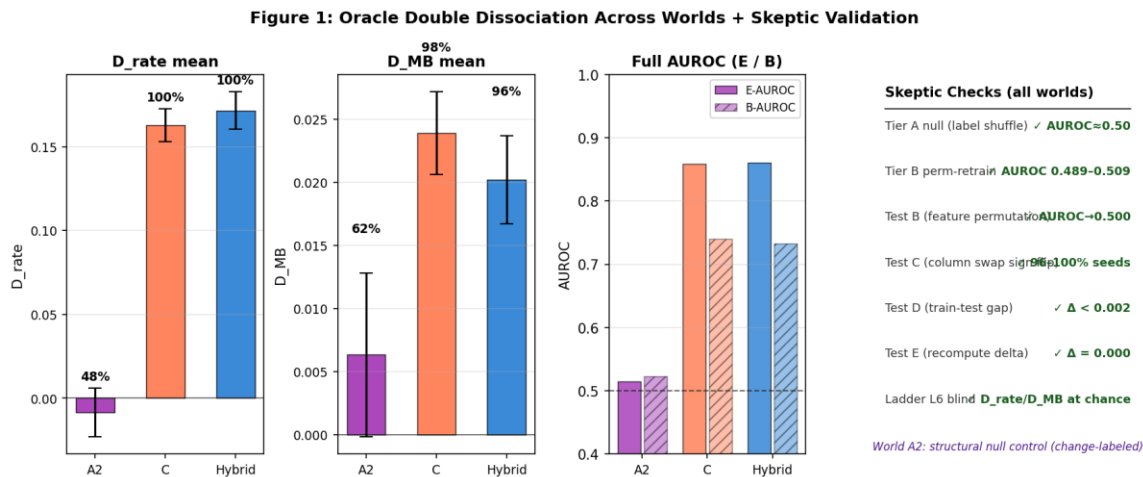


Figure 1: Oracle Double Dissociation + Skeptic Validation. Left three panels: D_{rate} , D_{MB} , and full AUROC (E/B) for Worlds A2, C, and Hybrid at oracle observability. Pass rates annotated above bars. Right panel: skeptic check summary (all pass). World A2 is near-chance throughout by construction.

Figure 2: Phase 1 Observability Ladder — Asymmetric Channel Robustness

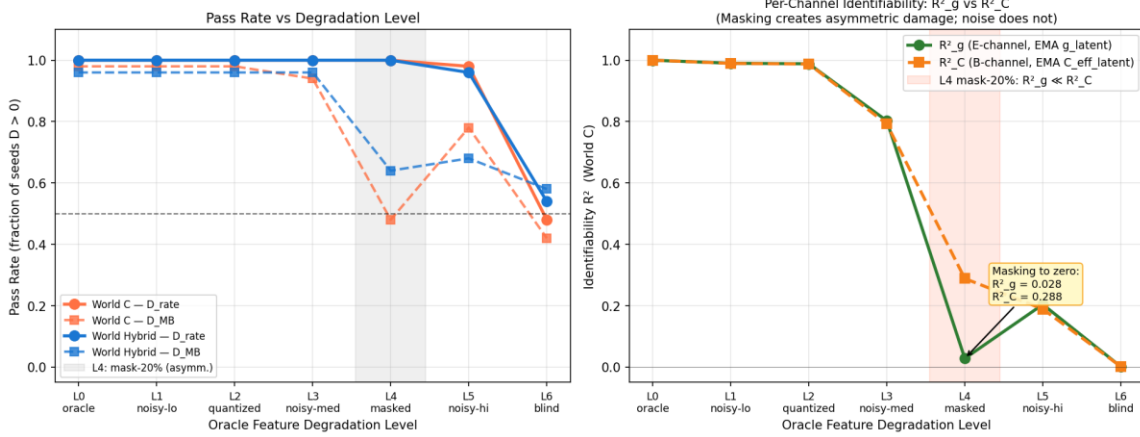


Figure 2: Phase 1 Oracle Degradation Ladder — Asymmetric Channel Damage. Left: D_{rate} and D_{MB} pass rate vs. degradation level for Worlds C and Hybrid. Right: Per-channel recoverability (R^2_g and R^2_C) for World C. The L4 zone (shaded) highlights the masking-induced asymmetry: $R^2_g = 0.028$ vs. $R^2_C = 0.288$.

Figure 3: Phase 2 Observability Ladder — Sensor Degradation + Decoder
(Left: World C recovers dissociation; Right: World Hybrid structural E-channel impossibility)

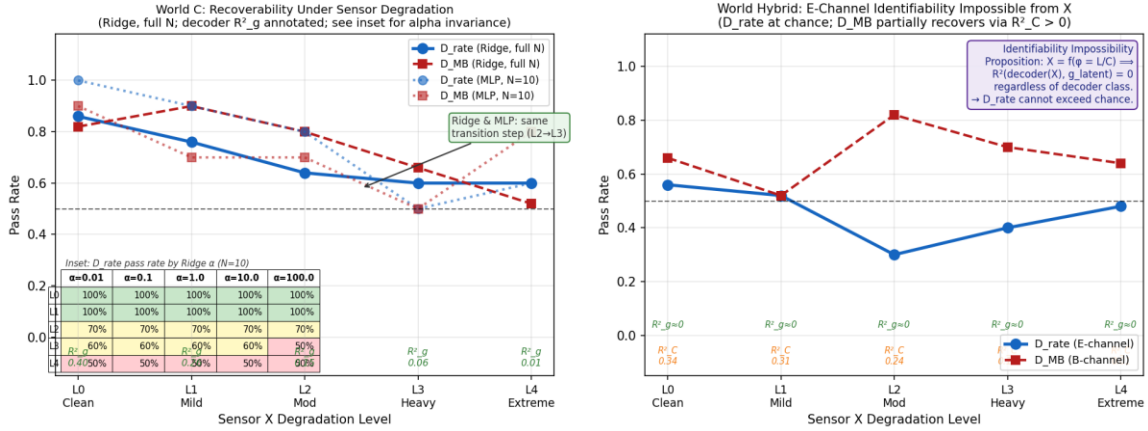


Figure 3: Phase 2 Sensor Degradation + Decoder. Left: World C pass rates for Ridge ($N = 50$) and MLP ($N = 3$, small confirmatory sample) decoders. Decoder R^2_g annotated per level. Inset table: Ridge alpha sweep pass rates ($N = 10$) — all five α values produce identical results, confirming hyperparameter invariance. Callout marks the shared Ridge/MLP transition zone ($L2 \rightarrow L3$). Right: World Hybrid — D_{rate} at chance across all levels ($R^2_g \approx 0$ throughout); annotation describes the empirical E-channel non-recoverability under tested decoders (Ridge and MLP), consistent with the dilution argument for the L/C observation operator.